

E2BCA1T1 – E – 25 – 2085



SECOND SEMESTER B.C.A. (REVISED NEP) DEGREE
EXAMINATION, JULY 2025
DATA STRUCTURES USING C (DSC – 1)
Theory

Time : 3 Hours]

[Max. Marks : 80

Instructions : 1) **All Sections are compulsory.**

2) **Read the instructions from each Section and answer accordingly.**

SECTION – A

Answer **all** the questions in this Section. **Each** carries **two** marks.

1. Define Structure.
2. What is pointer ? How to define and initialize pointer ?
3. Define primitive data structure.
4. What is searching and sorting ?
5. List the operations performed on Data structure.
6. Define stack.
7. What you mean by double ended queue ?
8. What is postfix expression ?
9. Write a pictorial representation of singly linked list.
10. What is complete binary tree ?

(10×2=20)

SECTION – B

Answer **any six** questions from this Section. **Each** carries **five** marks.

11. What is dynamic memory allocation ? Explain dynamic memory allocation functions with syntax.
12. Compare sequential search with Binary search technique.

[P.T.O.]



13. How to define structure ? Explain array as a structure and array with in structure with example.
14. Write an algorithm for merge sort.
15. Write a program to implement stack operation push and pop operation.
16. Explain circular queue with example.
17. What is linked list ? How to create singly linked list ? Explain with example.
18. Define Tree. Explain tree terminologies. (6×5=30)

SECTION – C

Answer **any three** questions from this Section. **Each** carries **10** marks.

19. Define Data structure. Explain classification of data structure with flowchart.
20. i) Write an algorithm for Binary search technique.
ii) Explain selection test technique with example. (5+5=10)
21. i) Write an algorithm to convert Infix and postfix expression.
ii) Convert the following expressions into prefix.
a) $(a+b)/c*d$ b) $a(*b-c)/d$ (6+4=10)
22. What is singly linked list ? Write an algorithm for singly linked list operation such as Insertion, Deletion and Searching. (3×10=30)